

## Company Database Example

```
mysql> create database Company;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use Company;  
Database changed
```

### Creating the table for given entities

```
mysql> CREATE TABLE Department (  
-> DepartmentID INT PRIMARY KEY,  
-> DepartmentName VARCHAR(100) NOT NULL,  
-> Location VARCHAR(100)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> desc Department;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| DepartmentID | int       | NO   | PRI | NULL    |      |  
| DepartmentName | varchar(100) | NO   |     | NULL    |      |  
| Location      | varchar(100) | YES  |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Employee (  
-> EmployeeID INT PRIMARY KEY,  
-> Name VARCHAR(100) NOT NULL,  
-> Email VARCHAR(100) UNIQUE,  
-> Phone VARCHAR(15),  
-> HireDate DATE,  
-> JobTitle VARCHAR(50),  
-> DepartmentID INT,  
-> ManagerID INT,  
-> FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID),  
-> FOREIGN KEY (ManagerID) REFERENCES Employee(EmployeeID)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> desc Employee;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| EmployeeID | int       | NO   | PRI | NULL    |      |  
| Name       | varchar(100) | NO   |     | NULL    |      |  
| Email      | varchar(100) | YES  | UNI | NULL    |      |  
| Phone      | varchar(15) | YES  |     | NULL    |      |  
| HireDate   | date       | YES  |     | NULL    |      |  
| JobTitle   | varchar(50) | YES  |     | NULL    |      |  
| DepartmentID | int       | YES  | MUL | NULL    |      |  
| ManagerID  | int       | YES  | MUL | NULL    |      |  
| Gender     | varchar(10) | YES  |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+  
9 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Project (
-> ProjectID INT PRIMARY KEY,
-> ProjectName VARCHAR(100) NOT NULL,
-> StartDate DATE,
-> EndDate DATE,
-> Budget DECIMAL(12,2),
-> DepartmentID INT,
-> FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> CREATE TABLE Assignment (
-> AssignmentID INT PRIMARY KEY,
-> EmployeeID INT,
-> ProjectID INT,
-> Role VARCHAR(50),
-> HoursWorked DECIMAL(5,2),
-> FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID),
-> FOREIGN KEY (ProjectID) REFERENCES Project(ProjectID)
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> desc Assignment;
```

| Field        | Type         | Null | Key | Default | Extra |
|--------------|--------------|------|-----|---------|-------|
| AssignmentID | int          | NO   | PRI | NULL    |       |
| EmployeeID   | int          | YES  | MUL | NULL    |       |
| ProjectID    | int          | YES  | MUL | NULL    |       |
| Role         | varchar(50)  | YES  |     | NULL    |       |
| HoursWorked  | decimal(5,2) | YES  |     | NULL    |       |

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE Salary (
-> SalaryID INT PRIMARY KEY,
-> EmployeeID INT,
-> Amount DECIMAL(10,2),
-> EffectiveDate DATE,
-> PayGrade VARCHAR(10),
-> FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> desc Salary;
```

| Field         | Type          | Null | Key | Default | Extra |
|---------------|---------------|------|-----|---------|-------|
| SalaryID      | int           | NO   | PRI | NULL    |       |
| EmployeeID    | int           | YES  | MUL | NULL    |       |
| Amount        | decimal(10,2) | YES  |     | NULL    |       |
| EffectiveDate | date          | YES  |     | NULL    |       |
| PayGrade      | varchar(10)   | YES  |     | NULL    |       |

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE Leaves (  
-> LeaveID INT PRIMARY KEY,  
-> EmployeeID INT,  
-> StartDate DATE,  
-> EndDate DATE,  
-> LeaveType VARCHAR(20),  
-> Status VARCHAR(20),  
-> FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> desc Leaves;
```

| Field      | Type        | Null | Key | Default | Extra |
|------------|-------------|------|-----|---------|-------|
| LeaveID    | int         | NO   | PRI | NULL    |       |
| EmployeeID | int         | YES  | MUL | NULL    |       |
| StartDate  | date        | YES  |     | NULL    |       |
| EndDate    | date        | YES  |     | NULL    |       |
| LeaveType  | varchar(20) | YES  |     | NULL    |       |
| Status     | varchar(20) | YES  |     | NULL    |       |

6 rows in set (0.00 sec)

## Alter commands(queries)

```
mysql> ALTER TABLE Employee ADD Gender VARCHAR(10);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Department;
+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| DepartmentID | int       | NO   | PRI | NULL    |      |
| DepartmentName | varchar(100) | NO   |     | NULL    |      |
| Location     | varchar(100) | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql> ALTER TABLE Employee MODIFY Phone VARCHAR(30);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Employee;
+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| EmployeeID | int       | NO   | PRI | NULL    |      |
| Name       | varchar(100) | NO   |     | NULL    |      |
| Email      | varchar(100) | YES  | UNI | NULL    |      |
| Phone      | varchar(30) | YES  |     | NULL    |      |
| HireDate   | date      | YES  |     | NULL    |      |
| JobTitle    | varchar(50) | YES  |     | NULL    |      |
| DepartmentID | int       | YES  | MUL | NULL    |      |
| ManagerID   | int       | YES  | MUL | NULL    |      |
| Gender      | varchar(10) | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+
9 rows in set (0.01 sec)
```

```
mysql> ALTER TABLE Department DROP COLUMN Location;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Department;
+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| DepartmentID | int       | NO   | PRI | NULL    |      |
| DepartmentName | varchar(100) | NO   |     | NULL    |      |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE Assignment ADD CONSTRAINT fk_assignment_employee
-> FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

mysql> desc Assignment;

| Field        | Type         | Null | Key | Default | Extra |
|--------------|--------------|------|-----|---------|-------|
| AssignmentID | int          | NO   | PRI | NULL    |       |
| EmployeeID   | int          | YES  | MUL | NULL    |       |
| ProjectID    | int          | YES  | MUL | NULL    |       |
| Role         | varchar(50)  | YES  |     | NULL    |       |
| HoursWorked  | decimal(5,2) | YES  |     | NULL    |       |

5 rows in set (0.00 sec)

mysql> ALTER TABLE Salary RENAME COLUMN Amount TO SalaryAmount;  
Query OK, 0 rows affected (0.03 sec)  
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Salary;

| Field         | Type          | Null | Key | Default | Extra |
|---------------|---------------|------|-----|---------|-------|
| SalaryID      | int           | NO   | PRI | NULL    |       |
| EmployeeID    | int           | YES  | MUL | NULL    |       |
| SalaryAmount  | decimal(10,2) | YES  |     | NULL    |       |
| EffectiveDate | date          | YES  |     | NULL    |       |
| PayGrade      | varchar(10)   | YES  |     | NULL    |       |

5 rows in set (0.00 sec)

mysql> RENAME TABLE Leaves TO EmployeeLeave;  
Query OK, 0 rows affected (0.04 sec)

mysql> desc EmployeeLeave;

| Field      | Type        | Null | Key | Default | Extra |
|------------|-------------|------|-----|---------|-------|
| LeaveID    | int         | NO   | PRI | NULL    |       |
| EmployeeID | int         | YES  | MUL | NULL    |       |
| StartDate  | date        | YES  |     | NULL    |       |
| EndDate    | date        | YES  |     | NULL    |       |
| LeaveType  | varchar(20) | YES  |     | NULL    |       |
| Status     | varchar(20) | YES  |     | NULL    |       |

6 rows in set (0.00 sec)

mysql> ALTER TABLE Assignment ADD CONSTRAINT chk\_hours\_positive CHECK (HoursWorked >= 0);  
Query OK, 0 rows affected (0.08 sec)  
Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc Assignment;
```

| Field        | Type         | Null | Key | Default | Extra |
|--------------|--------------|------|-----|---------|-------|
| AssignmentID | int          | NO   | PRI | NULL    |       |
| EmployeeID   | int          | YES  | MUL | NULL    |       |
| ProjectID    | int          | YES  | MUL | NULL    |       |
| Role         | varchar(50)  | YES  |     | NULL    |       |
| HoursWorked  | decimal(5,2) | YES  |     | NULL    |       |

```
5 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE Project DROP FOREIGN KEY project_ibfk_1;
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Project;
```

| Field        | Type          | Null | Key | Default | Extra |
|--------------|---------------|------|-----|---------|-------|
| ProjectID    | int           | NO   | PRI | NULL    |       |
| ProjectName  | varchar(100)  | NO   |     | NULL    |       |
| StartDate    | date          | YES  |     | NULL    |       |
| EndDate      | date          | YES  |     | NULL    |       |
| Budget       | decimal(12,2) | YES  |     | NULL    |       |
| DepartmentID | int           | YES  | MUL | NULL    |       |

```
6 rows in set (0.01 sec)
```

## Inserting the values

```
mysql> INSERT INTO Department (DepartmentID, DepartmentName) VALUES
```

```
-> (10, 'Research & Development'),
```

```
-> (11, 'Operations'),
```

```
-> (12, 'Finance');
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from Department;
```

```
+-----+-----+
```

```
| DepartmentID | DepartmentName |
```

```
+-----+-----+
```

```
|      10 | Research & Development |
```

```
|      11 | Operations |
```

```
|      12 | Finance |
```

```
+-----+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Employee (EmployeeID, Name, Email, Phone, HireDate, JobTitle, DepartmentID, ManagerID, Gender)
```

```
-> VALUES
```

```
-> (201, 'David Roy', 'david@company.com', '9111222333', '2020-05-20', 'Engineer', 10, NULL, 'Male'),
```

```
-> (202, 'Eva Thomas', 'eva@company.com', '9333444555', '2021-06-18', 'Coordinator', 11, 201, 'Female'),
```

```
-> (203, 'Frank Das', 'frank@company.com', '9444555666', '2023-02-10', 'Accountant', 12, 201, 'Male');
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from Employee;
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| EmployeeID | Name      | Email          | Phone   | HireDate | JobTitle      | DepartmentID | ManagerID | Gender |
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
|      201 | David Roy | david@company.com | 9111222333 | 2020-05-20 | Engineer      |      10 |      NULL | Male |
```

```
|      202 | Eva Thomas | eva@company.com   | 9333444555 | 2021-06-18 | Coordinator    |      11 |      201 | Female |
```

```
|      203 | Frank Das | frank@company.com | 9444555666 | 2023-02-10 | Accountant     |      12 |      201 | Male |
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Project (ProjectID, ProjectName, StartDate, EndDate, Budget, DepartmentID)
```

```
-> VALUES
```

```
-> (301, 'Neptune', '2024-03-01', '2024-12-01', 150000.00, 10),
```

```
-> (302, 'Hermes', '2024-04-10', '2024-11-30', 85000.00, 11),
```

```
-> (303, 'Poseidon', '2024-05-20', '2024-12-31', 65000.00, 12);
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from Project;
```

| ProjectID | ProjectName | StartDate  | EndDate    | Budget    | DepartmentID |
|-----------|-------------|------------|------------|-----------|--------------|
| 301       | Neptune     | 2024-03-01 | 2024-12-01 | 150000.00 | 10           |
| 302       | Hermes      | 2024-04-10 | 2024-11-30 | 85000.00  | 11           |
| 303       | Poseidon    | 2024-05-20 | 2024-12-31 | 65000.00  | 12           |

3 rows in set (0.00 sec)

```
mysql> INSERT INTO Assignment (AssignmentID, EmployeeID, ProjectID, Role, HoursWorked)
```

```
-> VALUES
```

```
-> (401, 201, 301, 'Tech Lead', 140.0),
```

```
-> (402, 202, 302, 'Ops Coordinator', 100.0),
```

```
-> (403, 203, 303, 'Finance Support', 55.0);
```

Query OK, 3 rows affected (0.02 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from Assignment;
```

| AssignmentID | EmployeeID | ProjectID | Role            | HoursWorked |
|--------------|------------|-----------|-----------------|-------------|
| 401          | 201        | 301       | Tech Lead       | 140.00      |
| 402          | 202        | 302       | Ops Coordinator | 100.00      |
| 403          | 203        | 303       | Finance Support | 55.00       |

3 rows in set (0.00 sec)

```
mysql> INSERT INTO Salary (SalaryID, EmployeeID, SalaryAmount, EffectiveDate, PayGrade)
```

```
-> VALUES
```

```
-> (501, 201, 90000.00, '2024-02-01', 'G7'),
```

```
-> (502, 202, 70000.00, '2024-02-01', 'G6'),
```

```
-> (503, 203, 60000.00, '2024-02-01', 'G5');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from Salary;
```

| SalaryID | EmployeeID | SalaryAmount | EffectiveDate | PayGrade |
|----------|------------|--------------|---------------|----------|
| 501      | 201        | 90000.00     | 2024-02-01    | G7       |
| 502      | 202        | 70000.00     | 2024-02-01    | G6       |
| 503      | 203        | 60000.00     | 2024-02-01    | G5       |

3 rows in set (0.00 sec)

```
mysql> INSERT INTO EmployeeLeave (LeaveID, EmployeeID, StartDate, EndDate, LeaveType, Status)
```

```
-> VALUES
```

```
-> (601, 201, '2024-07-10', '2024-07-14', 'Casual', 'Pending'),
```

```
-> (602, 202, '2024-08-01', '2024-08-03', 'Sick', 'Approved'),
```

```
-> (603, 203, '2024-09-05', '2024-09-15', 'Annual', 'Pending');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from EmployeeLeave;
+-----+-----+-----+-----+-----+-----+
| LeaveID | EmployeeID | StartDate | EndDate | LeaveType | Status |
+-----+-----+-----+-----+-----+-----+
| 601 | 201 | 2024-07-10 | 2024-07-14 | Casual | Pending |
| 602 | 202 | 2024-08-01 | 2024-08-03 | Sick | Approved |
| 603 | 203 | 2024-09-05 | 2024-09-15 | Annual | Pending |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

## Updating commands

```
mysql> UPDATE Employee SET Email = 'david.roy@corp.com', Phone = '9000111222' WHERE  
EmployeeID = 201;  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from Employee;
```

| EmployeeID | Name       | Email              | Phone      | HireDate   | JobTitle    | DepartmentID | ManagerID | Gender |
|------------|------------|--------------------|------------|------------|-------------|--------------|-----------|--------|
| 201        | David Roy  | david.roy@corp.com | 9000111222 | 2020-05-20 | Engineer    | 10           | NULL      | Male   |
| 202        | Eva Thomas | eva@company.com    | 9333444555 | 2021-06-18 | Coordinator | 11           | 201       | Female |
| 203        | Frank Das  | frank@company.com  | 9444555666 | 2023-02-10 | Accountant  | 12           | 201       | Male   |

3 rows in set (0.00 sec)

```
mysql> UPDATE Department SET DepartmentName = 'R&D Division' WHERE DepartmentID = 10;  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from Department;
```

| DepartmentID | DepartmentName |
|--------------|----------------|
| 10           | R&D Division   |
| 11           | Operations     |
| 12           | Finance        |

3 rows in set (0.00 sec)

```
mysql> UPDATE Project SET Budget = 160000.00, EndDate = '2025-02-28' WHERE ProjectID = 301;  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from Project;
```

| ProjectID | ProjectName | StartDate  | EndDate    | Budget    | DepartmentID |
|-----------|-------------|------------|------------|-----------|--------------|
| 301       | Neptune     | 2024-03-01 | 2025-02-28 | 160000.00 | 10           |
| 302       | Hermes      | 2024-04-10 | 2024-11-30 | 85000.00  | 11           |
| 303       | Poseidon    | 2024-05-20 | 2024-12-31 | 65000.00  | 12           |

3 rows in set (0.00 sec)

```
mysql> UPDATE Assignment SET Role = 'Senior Tech Lead', HoursWorked = 150.0 WHERE  
AssignmentID = 401;  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

mysql> select \* from Assignment;

| AssignmentID | EmployeeID | ProjectID | Role             | HoursWorked |
|--------------|------------|-----------|------------------|-------------|
| 401          | 201        | 301       | Senior Tech Lead | 150.00      |
| 402          | 202        | 302       | Ops Coordinator  | 100.00      |
| 403          | 203        | 303       | Finance Support  | 55.00       |

3 rows in set (0.00 sec)

mysql> UPDATE Salary SET SalaryAmount = 95000.00, PayGrade = 'G8' WHERE SalaryID = 501;  
Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select \* from Salary;

| SalaryID | EmployeeID | SalaryAmount | EffectiveDate | PayGrade |
|----------|------------|--------------|---------------|----------|
| 501      | 201        | 95000.00     | 2024-02-01    | G8       |
| 502      | 202        | 70000.00     | 2024-02-01    | G6       |
| 503      | 203        | 60000.00     | 2024-02-01    | G5       |

3 rows in set (0.00 sec)

mysql> UPDATE EmployeeLeave SET Status = 'Approved' WHERE LeaveID = 601;  
Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select \* from EmployeeLeave;

| LeaveID | EmployeeID | StartDate  | EndDate    | LeaveType | Status   |
|---------|------------|------------|------------|-----------|----------|
| 601     | 201        | 2024-07-10 | 2024-07-14 | Casual    | Approved |
| 602     | 202        | 2024-08-01 | 2024-08-03 | Sick      | Approved |
| 603     | 203        | 2024-09-05 | 2024-09-15 | Annual    | Pending  |

3 rows in set (0.01 sec)

mysql> UPDATE Employee SET DepartmentID = 10;  
Query OK, 2 rows affected (0.01 sec)

Rows matched: 3 Changed: 2 Warnings: 0

mysql> select \* from Employee;

| EmployeeID | Name       | Email              | Phone      | HireDate   | JobTitle    | DepartmentID | ManagerID | Gender |
|------------|------------|--------------------|------------|------------|-------------|--------------|-----------|--------|
| 201        | David Roy  | david.roy@corp.com | 9000111222 | 2020-05-20 | Engineer    | 10           | NULL      | Male   |
| 202        | Eva Thomas | eva@company.com    | 9333444555 | 2021-06-18 | Coordinator | 10           | 201       | Female |
| 203        | Frank Das  | frank@company.com  | 9444555666 | 2023-02-10 | Accountant  | 10           | 201       | Male   |

3 rows in set (0.00 sec)